

$\mathcal{K}_{0^+}^{\#1}$
 $\mathcal{K}_{0^-}^{\#1}$
 $\alpha\beta\chi$

$\mathcal{K}_{0^+}^{\#1} \dagger$
 $-\frac{3\Upsilon_1 k^2}{16}$
0

$\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$
0
0

$\mathcal{K}_{1^+}^{\#1}$
 $\mathcal{K}_{1^+}^{\#2}$
 $\mathcal{K}_{1^-}^{\#1}$
 $\mathcal{K}_{1^-}^{\#2}$

$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$
 $-\frac{\Upsilon_2 k^2}{16}$
 $\frac{\Upsilon_2 k^2}{8\sqrt{2}}$
0
0

$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$
 $\frac{\Upsilon_2 k^2}{8\sqrt{2}}$
 $-\frac{\Upsilon_2 k^2}{8}$
0
0

$\mathcal{K}_{1^-}^{\#1} \dagger^\alpha$
0
0
0
0

$\mathcal{K}_{1^-}^{\#2} \dagger^\alpha$
0
0
0
0

 $\mathcal{K}_{2^+}^{\#1}$
 $\mathcal{K}_{2^-}^{\#1}$
 $\alpha\beta\chi$
 $\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$
 $\frac{3\Upsilon_3 k^2}{16}$
0

 $\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$
0
 $\frac{3k^2 \kappa_{15}^{(4)}}{2}$

$\mathcal{J}_{0^+}^{\#1}$
 $\mathcal{J}_{0^-}^{\#1}$
 $\alpha\beta\chi$

$\mathcal{J}_{0^+}^{\#1} \dagger$
 $\frac{1}{\text{Det}(0^+)}$
0

$\mathcal{J}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$
0
0

$\mathcal{J}_{1^+}^{\#1}$
 $\mathcal{J}_{1^+}^{\#2}$
 $\mathcal{J}_{1^-}^{\#1}$
 $\mathcal{J}_{1^-}^{\#2}$

$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$
 $\frac{1}{3\text{Det}(1^+)}$
 $-\frac{\sqrt{2}}{3\text{Det}(1^+)}$
0
0

$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$
 $-\frac{\sqrt{2}}{3\text{Det}(1^+)}$
 $\frac{2}{3\text{Det}(1^+)}$
0
0

$\mathcal{J}_{1^-}^{\#1} \dagger^\alpha$
0
0
0
0

$\mathcal{J}_{1^-}^{\#2} \dagger^\alpha$
0
0
0
0

 $\mathcal{J}_{2^+}^{\#1}$
 $\mathcal{J}_{2^-}^{\#1}$
 $\alpha\beta\chi$
 $\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$
 $\frac{1}{\text{Det}(2^+)}$
0

 $\mathcal{J}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$
0
 $\frac{2}{3k^2 \kappa_{15}^{(4)}}$

Abbreviations used in matrices

$\Upsilon_1 == 8 \kappa_{1}^{(4)} + \kappa_{15}^{(4)} + 6 \kappa_{3}^{(4)}$
 $\&\& \Upsilon_2 == \kappa_{15}^{(4)} - 2 \kappa_{3}^{(4)}$
 $\&\&$

$\Upsilon_3 == 11 \kappa_{15}^{(4)} - 6 \kappa_{3}^{(4)}$
 $\&\& \text{Det}(0^+) == -\frac{3}{16} k^2 (8 \kappa_{1}^{(4)} + \kappa_{15}^{(4)} + 6 \kappa_{3}^{(4)})$
 $\&\&$

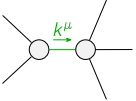
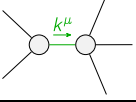
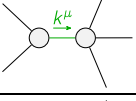
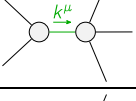
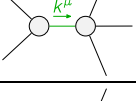
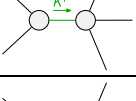
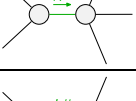
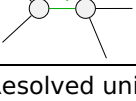
$\text{Det}(1^+) == -\frac{3}{16} k^2 (\kappa_{15}^{(4)} - 2 \kappa_{3}^{(4)})$
 $\&\& \text{Det}(2^+) == \frac{3}{16} k^2 (11 \kappa_{15}^{(4)} - 6 \kappa_{3}^{(4)})$

Lagrangian

$\kappa_{1}^{(4)} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + \frac{3}{2} \kappa_{15}^{(4)} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + \kappa_{3}^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \frac{9}{4} \kappa_{15}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} -$

$\frac{1}{2} \kappa_{3}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} - \frac{9}{8} \kappa_{15}^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} + \frac{5}{4} \kappa_{3}^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} - 3 \kappa_{15}^{(4)} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} -$

$\frac{9}{8} \kappa_{15}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \frac{1}{4} \kappa_{3}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \kappa_{15}^{(4)} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} + \kappa_{15}^{(4)} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_{\delta} \partial^{\alpha} \mathcal{J}^{\beta\chi\delta} + \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\delta\beta\chi} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\alpha\beta\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\delta\alpha\beta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_{\delta} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\alpha\chi\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\delta\alpha\chi} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\beta\alpha\delta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\alpha\beta\chi} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_{\chi} \partial_{\beta} \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_1^{\#1\alpha} == 0$	3	$\partial_{\chi} \partial^{\alpha} \mathcal{J}^{\beta}_{\beta}{}^{\chi} + \partial_{\chi} \partial^{\chi} \mathcal{J}^{\beta\alpha}_{\beta} == \partial_{\chi} \partial_{\beta} \mathcal{J}^{\beta\alpha\chi}$	
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_{\chi} \mathcal{J}^{\alpha\beta\chi} + \partial_{\chi} \mathcal{J}^{\chi\alpha\beta} == \partial_{\chi} \mathcal{J}^{\beta\alpha\chi}$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{7 \kappa_{15}^{(4)} + 2 \kappa_3^{(4)}}{\kappa_{15}^{(4)} - 2 \kappa_{15}^{(4)} \kappa_3^{(4)}}$
	1	0	$\frac{-57 \kappa_{15}^{(4)2} + 20 \kappa_{15}^{(4)} \kappa_3^{(4)} - 4 \kappa_3^{(4)2}}{\kappa_{15}^{(4)} (11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}) (\kappa_{15}^{(4)} - 2 \kappa_3^{(4)})}$
	1	0	$\frac{19 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}}{11 \kappa_{15}^{(4)2} - 6 \kappa_{15}^{(4)} \kappa_3^{(4)}}$
	1	0	$\frac{-97 \kappa_{15}^{(4)2} + 116 \kappa_{15}^{(4)} \kappa_3^{(4)} - 36 \kappa_3^{(4)2}}{\kappa_{15}^{(4)} (11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}) (\kappa_{15}^{(4)} - 2 \kappa_3^{(4)})}$
	1	0	$\frac{34 \kappa_{15}^{(4)2} + 40 \kappa_{15}^{(4)} \kappa_3^{(4)} - 24 \kappa_3^{(4)2} - \sqrt{5545 \kappa_{15}^{(4)4} - 8584 \kappa_{15}^{(4)3} \kappa_3^{(4)} + 4696 \kappa_{15}^{(4)2} \kappa_3^{(4)2} - 1056 \kappa_{15}^{(4)} \kappa_3^{(4)3} + 144 \kappa_3^{(4)4}}}{11 \kappa_{15}^{(4)3} - 28 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 12 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
	1	0	$\frac{34 \kappa_{15}^{(4)2} + 40 \kappa_{15}^{(4)} \kappa_3^{(4)} - 24 \kappa_3^{(4)2} + \sqrt{5545 \kappa_{15}^{(4)4} - 8584 \kappa_{15}^{(4)3} \kappa_3^{(4)} + 4696 \kappa_{15}^{(4)2} \kappa_3^{(4)2} - 1056 \kappa_{15}^{(4)} \kappa_3^{(4)3} + 144 \kappa_3^{(4)4}}}{11 \kappa_{15}^{(4)3} - 28 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 12 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
	1	0	$-\frac{-8 \kappa_{15}^{(4)2} + 16 \kappa_{15}^{(4)} \kappa_3^{(4)} + \sqrt{(\kappa_{15}^{(4)} - 2 \kappa_3^{(4)})^2 (313 \kappa_{15}^{(4)2} - 132 \kappa_{15}^{(4)} \kappa_3^{(4)} + 36 \kappa_3^{(4)2})}}{\kappa_{15}^{(4)} (11 \kappa_{15}^{(4)} - 6 \kappa_3^{(4)}) (\kappa_{15}^{(4)} - 2 \kappa_3^{(4)})}$
	1	0	$\frac{8 \kappa_{15}^{(4)2} - 16 \kappa_{15}^{(4)} \kappa_3^{(4)} + \sqrt{(\kappa_{15}^{(4)} - 2 \kappa_3^{(4)})^2 (313 \kappa_{15}^{(4)2} - 132 \kappa_{15}^{(4)} \kappa_3^{(4)} + 36 \kappa_3^{(4)2})}}{11 \kappa_{15}^{(4)3} - 28 \kappa_{15}^{(4)2} \kappa_3^{(4)} + 12 \kappa_{15}^{(4)} \kappa_3^{(4)2}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		